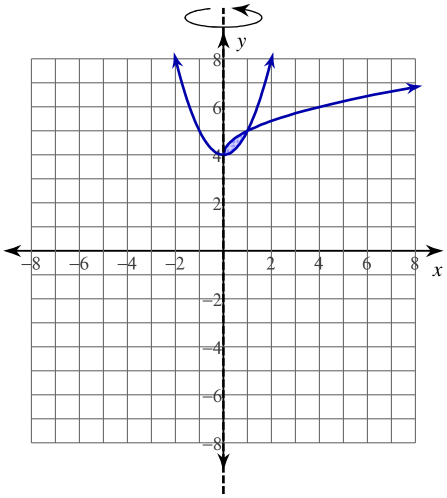
Name________________________________

Volumes by Cylindrical Shells

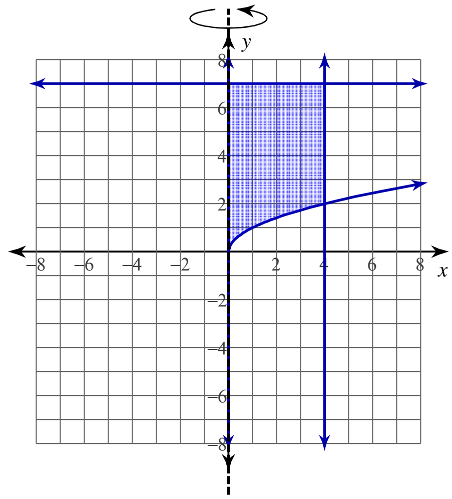
Date________________ Period____

For each problem, use the method of cylindrical shells to find the volume of the solid that results when the region enclosed by the curves is revolved about the *y*-axis.

1) $y = \sqrt{x} + 4$
$y = x^2 + 4$



2) $y = 7$
$y = \sqrt{x}$
$x = 0$
$x = 4$



Critical thinking question:

3) Solve problem 2 using the method of washers. Why is this problem easier using cylindrical shells?